Soil Nutrient Analytics API – Documentation / User Manual

Version: 1.0

Provider: *TENSOR / Peru – Coffee and Sugarcane Crop Monitoring*

Underlying Technology: AWS Lambda, API Gateway, S3, Regression Algorithms

Purpose: Retrieve soil nutrient statistics and nutrient visualization maps for a given farmland.

1. Overview

The **Soil Nutrient Analytics API** enables clients to retrieve for each farmland over a bare soil period (for eg. Aug-2025):

- **Statistical summaries** (min, max, mean, median, std)
- **Geospatially clipped nutrient values** from high-resolution raster files
- **Interactive HTML-based nutrient maps** (with satellite basemap & colorbar)

This service processes geospatial data stored in **AWS S3** using a **serverless Lambda function**.

The API is designed for:

- Agriculture analytics platforms
- Crop monitoring applications
- Precision farming dashboards

2. API Endpoint

GET Method

https://<api-id>.execute-api.<region>.amazonaws.com/prod/nutrients

Example:

https://7yhb1ez8y7.execute-api.ap-south-1.amazonaws.com/S1/nutrients?farmland_id=NORCAJ2338&month=Aug-2025

3. Input Parameters

The API accepts **two query parameters**.

Parameter	Type	Required	Example	Description
farmland_id	string	Yes	NORCAJ2338	Unique ID of farmland polygon stored in S3 (via kml file)

month	string	Yes	Aug-2025	Folder name containing nutrient GeoTIFFs for the month

Notes:

- farmland_id must match an entry as per the farmland locations file shared by client.
- month must match a folder like Aug-2025/, Oct-2025/, etc.
- Each month folder contains one or more **Cloud-Optimized GeoTIFF (COG)** nutrient rasters.

Example Request

GET /prod/nutrients?farmland_id=NORCAJ2338&month=Aug-2025

4. Output Structure (JSON Response)

The API returns a structured JSON response with the following keys:

4.1 Top-Level Structure

```
{
  "status": "success",
  "farmland_id": "NORCAJ2338",
  "month": "Oct-2025",
  "nutrients": { ... },
  "errors": { ... }
}
```

4.2 Nutrient Statistics Format

For each nutrient raster file (COG), the system computes:

- Minimum value
- Maximum value
- Mean value
- Median value
- Standard deviation
- Interactive HTML visualization map

Example nutrient output:

```
"nutrients": {
  "pH": {
    "min": 5.21,
```

```
    "max": 7.89,
    "mean": 6.72,
    "median": 6.70,
    "std": 0.31,
    "map_html": "<html>...base64 encoded folium map...</html>"
  }
}
```


map_html field

This is a **complete interactive map** containing:

- Satellite basemap
- Clipped nutrient raster overlay
- Nutrient legend + colorbar
- Farmland boundary
- Zoom & pan controls

Clients can directly embed this HTML in:

- Web dashboards
- Mobile apps (WebView)
- React / Angular / Vue applications

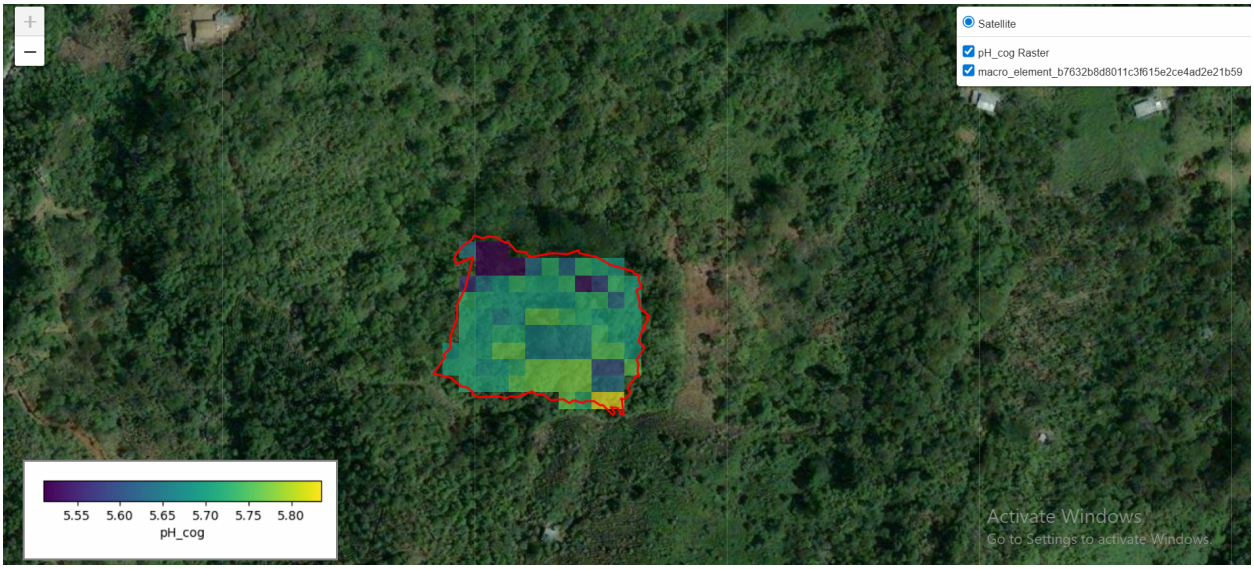


Figure . Interactive pH distribution map for the selected farmland, generated by the Soil Nutrient Analytics API. The map displays clipped pH values overlaid on satellite imagery, with a color-scaled legend indicating the nutrient intensity range.

4.3 Error Field Format

If a nutrient raster fails processing:

"errors": {
  "pH": "Failed to compute stats or map for Oct-2025/pH_cog.tif: <error-message>"
}
If everything succeeds:
"errors": {}

5. Error Conditions and Responses

Condition	Example Message
Missing farmland ID	`{"status": "error", "message": "Missing required field: farmland_id"}`
Missing month	`{"status": "error", "message": "Missing required field: month"}`
Farmland ID not found	`{"status": "error", "message": "Farmland ID 'NORCAJ23389' not found in KML", "details": "Verify farmland_id or update the KML file"}` in KML"
Month folder missing	`{"status": "error", "message": "Month folder 'Oct-2025' not found in S3 bucket 'peru-soil'", "details": "Ensure folder exists: e.g., 'Aug-2025/'"}`

6. Internal Processing Flow (Client Understanding)

The system provides soil nutrient analytics by leveraging **trained regression models**, **bare-soil satellite observations**, and **cloud-optimized geospatial data processing**.

Final API Response

The final output returned to the client includes:

- Farmland ID
- Month requested
- Nutrient-wise statistics
- Interactive HTML map for each nutrient
- Error log (if any nutrient layer fails)

This allows seamless integration of soil nutrient analytics into external applications, dashboards, and mobile/web interfaces.

7. Client Integration Guide

7.1 Direct Browser Call

https://.../prod/nutrients?farmland_id=XXX&month=YYY

7.2 JavaScript Fetch

```
fetch("https://.../nutrients?farmland_id=NORCAJ2338&month=Oct-2025")
  .then(r => r.json())
  .then(data => {
     document.getElementById("map").innerHTML = data.nutrients.pH.map_html;
  });
```

7.3 Python Request

```
import requests
url = "https://.../nutrients"
params = {"farmland_id": "NORCAJ2338", "month": "Oct-2025"}
res = requests.get(url, params=params)
data = res.json()
print(data["nutrients"]["pH"]["mean"])
```

7.4 Mobile App / WebView

Using the map_html content is straightforward:

```
WebView(
  initialUrl: Uri.dataFromString(data["nutrients"]["pH"]["map_html"],
    mimeType: 'text/html'
  ).toString(),
)
```

8. Security

- API Gateway calls Lambda using IAM permissions
- Lambda pulls S3 data using the Lambda Execution Role
- No AWS keys are exposed to clients
- Client sees only the HTTPS endpoint
- Optional: **Use API Keys + Usage Plans + Throttling + IP Controls (optional)**

9. Versioning and Updates

Future enhancements may include:

- Additional nutrient indices
- Multi-month aggregation (if required; if there is a continuous longer bare soil period)
- Farmland time series trends
- AI-based anomaly detection

All updates will be backward-compatible unless explicitly communicated.